

Ben Sirivallop

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EDUCATION

University of California, Irvine

Bachelor of Science in Computer Science (Expected June 2027), Minor in Economics

Sep. 2023 – Present

Irvine, CA

EXPERIENCE

ProGroup

Software Engineer Intern

Dec. 2025 – Present

Irvine, CA

- Reduced costs **80%** by migrating a PowerApps tool to **TypeScript, Node.js, MySQL, Redis, and AWS**.
- Integrated **Redis caching** on key endpoints, cutting average request times by **25%** and reducing backend load.
- Deployed to **AWS ECS** via **CI/CD pipelines** and implemented notifications with **AWS SQS and Lambda**.
- Achieved **83% unit test coverage** using **Jest and Supertest** to ensure high API reliability and stability.

Thai Oil

Data Science Intern

Jun. 2024 – Sep. 2024

Bangkok, Thailand

- Automated diagnostic workflows in **Python**, reducing inspection time by **60%** and increasing productivity.
- Built a data pipeline processing **40GB+** of IoT sensor data from **AWS S3** to **SageMaker** for ML training.

PROJECTS

Squadwork

TypeScript, Node.js, Redis, WebSockets, JWT, Docker, AWS ECS

Jun. 2025 – Present

- Integrated a **Redis caching layer**, reducing **API response times** by **85%** and decreasing **database load**.
- Automated **CI/CD pipelines** on **AWS ECS**, cutting **deployment time** by **80%** (**15 minutes** to **3 minutes**).
- Architected **RESTful APIs** with **JWT-based authentication**, implementing rotating **refresh** and **access token** cycles for secure, **stateless session management**.
- Built **real-time team chat** using **WebSockets** with **Redis-backed message persistence** and delivery.

Fabflix

Java, MySQL, Tomcat, Kubernetes, Docker, AWS, Distributed Systems

Jan. 2026 – Mar. 2026

- Deployed **microservices** in **Docker** containers on a **Kubernetes cluster (AWS EKS)** running on **AWS EC2**.
- Implemented **MySQL master-slave replication** and **load balancing** across a cluster of **Tomcat** instances.
- Designed a relational **MySQL database schema** modeling complex relationships between movies, stars, genres, and user data, with all interactions secured via **PreparedStatements** to prevent **SQL injection**.

AI Native Document Editor

TypeScript, Node.js, OpenAI API, WebSockets, PostgreSQL

Oct. 2025 – Dec. 2025

- Designed a **user access and permissions system** to manage read/write permissions for multiple users.
- Built a text-editor with **live collaboration** using **WebSockets**, enabling users to edit documents simultaneously.

TECHNICAL SKILLS

Languages: TypeScript, SQL, JavaScript, HTML/CSS, Python, Java, C++

Frameworks & Libraries: React, Node.js, Express, Next.js, FastAPI, Flask, Prisma, Tomcat

Databases & Caching: Redis, PostgreSQL, MySQL

Cloud & DevOps: AWS (ECS, EC2, CI/CD, Docker, Kubernetes, EKS, Lambda, S3), Git, GCP, GitHub Actions

Testing & Tools: JWT, Jest, Supertest, OAuth 2.0